

Project Title - IlliniSpaces

Project Summary

IlliniSpaces is a platform designed to help students and organizations efficiently locate and claim available meeting rooms on campus. By integrating university scheduling data from `courses.illinois.edu` and `one.illinois.edu`, the platform provides real-time insights into open spaces. Users can filter rooms based on availability, capacity, and distance, streamlining the booking process. Users can also make informal reservations on our platform, allowing for user coordination based on room availability. Additionally, an interactive campus map will enhance visualization by displaying room availability through color-coded indicators, with potential expansions for route mapping and calendar-based scheduling. With a centralized and accessible interface, IlliniSpaces optimizes campus space usage and enhances the student experience at the University of Illinois Urbana-Champaign.

Description

IlliniSpaces simplifies the challenge of finding campus meeting spaces by aggregating university scheduling data into a user-friendly and visually interactive platform. The application pulls real-time availability from `courses.illinois.edu` (classroom usage) and `one.illinois.edu` (event reservations) to provide an up-to-date list of open rooms. Users can search these rooms and filter their search by availability, room capacity, and proximity. Users will also be able to create informal room reservations on our platform, and update/delete these room reservations, with these reservations able to be viewed (read) by other users. In addition, we will attempt to add UIUC Google OAuth and only allow registration by `illinois.edu` emails. This will ensure that only registered users can make informal reservations.

To further enhance usability, we plan to implement an interactive campus map using Leaflet.js and OpenStreetMap Tiles, allowing users to visualize available rooms within buildings through color-coded indicators. A future iteration may also include highlighted walking directions and a room-specific availability calendar, leveraging a tool like React Calendar for better scheduling insights. Future expansions may also support Registered Student Organization (RSO) bookings, increasing accuracy for student group meetings. By consolidating campus-wide room availability, IlliniSpaces reduces scheduling inefficiencies and fosters student collaboration at UIUC.

Creative Component

Aside from our search for rooms by name, availability, capacity, and distance, one creative component for us to implement would be an interactive map of UIUC's campus that details the information for each room and allows you to visualize the availability of certain locations (i.e. buildings) using colors. This could become a complex feature to implement, and we will likely attempt a simple version of this to demonstrate the potential uses of this interface. If time and resources permit, we could also generate highlighted street directions to the location

directly on our map. Note that this would not be a Google Maps API call or embed, but rather it would be a custom map implementation using a library such as Leaflet.js and OpenStreetMap Tiles. We could also build a feature that displays a calendar of some sort for each room that shows its future availability. This would require data pulling, and populating a well designed calendar such as React Calendar on the frontend.

Usefulness

Many smaller student groups have a difficult time trying to find a meeting spot. Often, they will go to a smaller room in Grainger, Siebel, CIF, etc. only to then get kicked out by a different group or class that had the spot reserved. Our app will alleviate this issue by allowing everyone to have one centralized platform to find spots that are open. Users of this website will be able to search for different rooms and availability by filtering based on their needs, such as the size of the room or its location. Our application will have real time updates of new events and/or courses and will update the availability of rooms. There already exists a similar platform for this purpose that many larger RSOs use affiliated with the Student Org Development & Administration (SODA) office. SODA provides a private online booking system. However, through our research, we have found that many have problems with the filtering options allowed by the SODA system and its ease of use. The bigger problem is that most students can't access it, which we hope to fix through our project. *IlliniSpaces* will be a better option due to the UI being more streamlined, allowing the public to access it, allowing more filtering and search options (e.g. location), and allowing people to informally reserve rooms.

Realness

There are several different possible data sources for us to use in this project. We will need datasets to obtain information about reservable spaces and classrooms, as well as current classes and events going on. Here are some possible data sources to obtain this information:

- courses.illinois.edu - Course information, times, locations
- one.illinois.edu - Events, certain room reservations, and times
- calendars.illinois.edu - Events and times
- <https://operations.illinois.edu/facility-scheduling-and-resources/event-planning-resources/master-space-contact-list/> - Information about specific non classroom spaces
- https://operations.illinois.edu/wp-content/uploads/2025/02/FINALSpringSchedule-January21_May15_2025_0205.pdf - Campus building hours.
- <https://operations.illinois.edu/facility-scheduling-and-resources/reservable-spaces/>
- <https://operations.illinois.edu/wp-content/uploads/2025/01/Spring25ResvRooms-Jan-15-Capacity-Updated-1.xlsx> - Gives reservable room restrictions and capacities

Some of these data sources will likely require HTML scraping to obtain the information, after which we will likely store it as csv data. Additionally, some of the sources come as Google sheets or PDFs which we will parse using tools to extract any useful information about classes or rooms.

A secondary data source will be the Open Route Service Matrix Service API which will take in a set of destination coordinates and origin coordinates and can find pairwise distances (both many to one and many to many) between points for various transportation modes. In addition, we will need a way to map building names to coordinates, which we will do using OpenStreetMap's Nominatim API.

Functionality Description

A detailed description of the **functionality** that your website offers. This is where you talk about what the website delivers. Talk about how a user would interact with the application (i.e., things that one could create, delete, update, or search for). **Read the requirements for stage 4 to see what other functionalities you want to provide to the users.**

- **Space Searching & Reservation**

- Users can search available spaces on campus that satisfy a list of requirements they can customize, including type of classroom, capacity, time availability, features, and distance.
 - This will be a stored procedure that includes an INTERSECT operation where the available Spaces is the intersection of rooms not occupied by classes and the rooms not yet reserved by users.
- Users can search available spaces within their **favorited buildings**.
- Users can add and remove their **favorited buildings**. (create, delete)
- Users can check their reservations (read).
- Users can check availability status of arbitrary spaces (read)
- Users can change their reservations (update)
 - Constraint that each user can only reserve one Space for each time period.
 - Constraint that no overlapping reservations are allowed for one Space

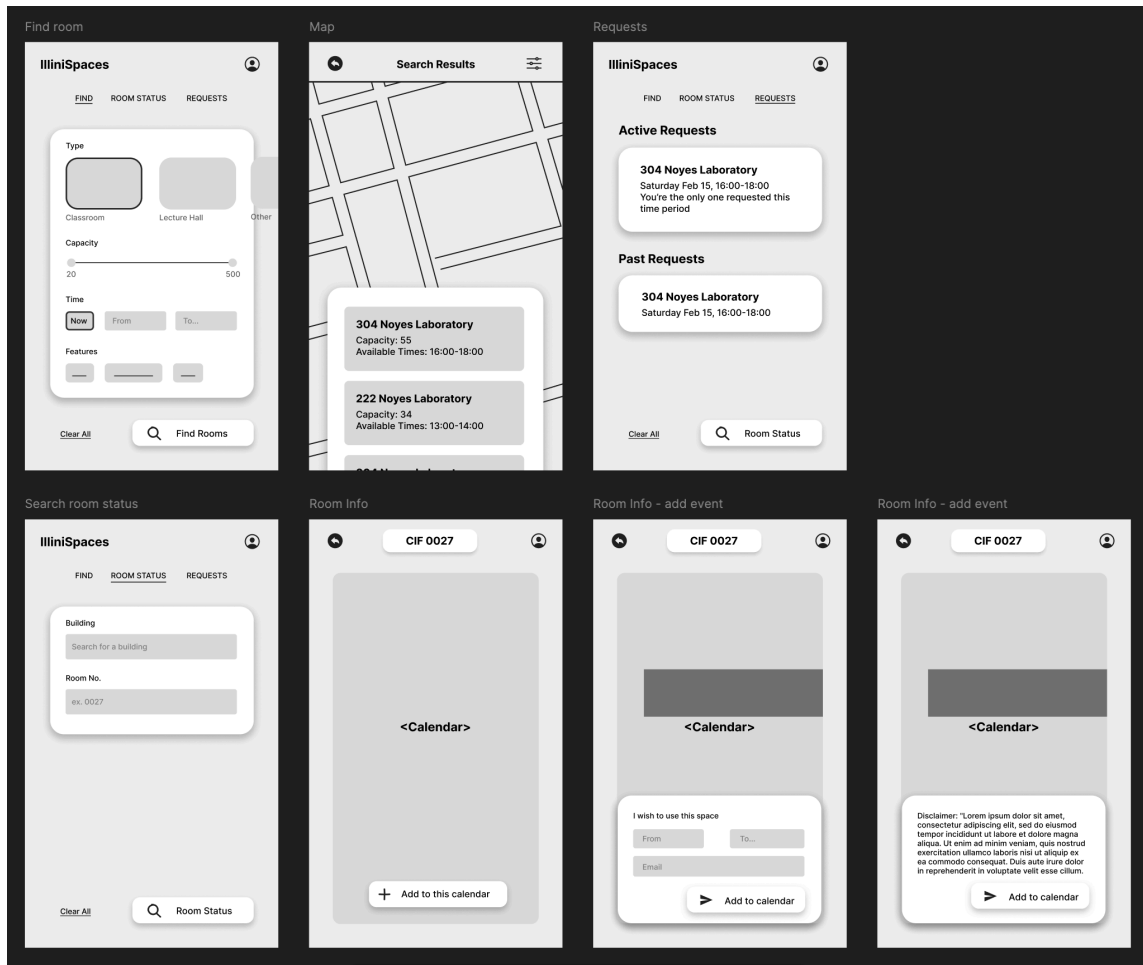
- **Navigation**

- Users can see nearby available spaces on a map. This will include a GROUP BY aggregation to group available rooms by buildings, and an INTERSECT operation (same as searching)

- **User Login**

- Includes illinois email, custom password

A low-fidelity UI mockup: What do you imagine your final application's interface might look like? A PowerPoint slide or a pencil sketch on a piece of paper works!



■ **Project work distribution:** Who will be responsible for each of the tasks or subtasks?

Explain how backend systems will be distributed across members. Be as specific as possible as this could be part of the final peer evaluation metrics.

- Roy - Web scraping and data gathering for rooms, events, capacities, classes, times, buildings, etc. Modeling the data and consolidating it into tables. Possibly automating this process to periodically update tables.
- Nehan - Integrating OSM Nominatim API and openroute service Matrix Service API into our application, and then using building coordinates, distances, and developing our interactive map frontend using Leaflet.
- Atsi - Adding available rooms on top of our interactive map buildings, room data querying, CRUD for room reservations and associated frontend UI for logged in users.
- Somrishi - Searching, filtering, user login screen and UIUC (Google?) OAuth enforcement.